Week 2 Tutorial Notes

▼ Agenda

- Q4 → after ‘Data Modelling’
- Q7 … relatively straightforward ER data modelling exercise
- Q8 → after discussing multivalued attributes
- Q9 … different relationships in ER
- Q11 → first ER for class to try
- Q15 … a substantial ER data modelling exercise (if you have time)

Pre-tute

- Introductions 🙂
- Discourse!!
- Quiz due Friday

Course Outline

Content

- Topic Videos
- Lectures
- Textbooks (OPTIONAL)

Activities

- Tutorials
- Prac exercises
- **Quizzes**
- **Assignments**

- **Exam**

▼ Mark Distribution

Item	Topics	Due	Marks	Contributes to
Quizzes	All topics	Weeks 2,3,4,7,8,10	12%	1,2,3,4,5,6,7,8
Assignment 1	SQL/PLpgSQL	Week 5	13%	3,4
Assignment 2	Python/SQL	Week 10	15%	5
Final Exam	All topics	Exam period	60%	1,2,3,4,5,6,7,8

Contact

- Administrative
  - cs3311@cse.unsw.edu.au
- Technical / Coursework
  - Discourse!!
- If you're unsure, contact me:
  - z5419507@ad.unsw.edu.au

Setup

1. Access the vxdb02 server
   - on cse server:

     ```
     $ ssh vxdb02
     ```

   - else:

     ```
     $ ssh YourUserName@vxdb02.cse.unsw.edu.au
     ```

2. (If first time running)

   ```
   $ 3311 pginit
   ```

3. Tell vxdb02 the source of the psql server and start the server

```
$ source /localstorage/$USER/env
$ p1
```

4. Work with a specific database

```
$ psql SomeDatabase
```

5. Stop PSQL server

```
$ p0
```

After this, look into ‘help’ command

Data Modelling

Aims to:

- describe what **data** is contained in the database

  (e.g., entities: students, courses, accounts, branches, patients, ...)

- describe **relationships** between data items / objects

  (e.g., John is enrolled in COMP3311, Tom's account is held at Coogee)

- describes **constraints** on data

  (e.g., 7-digit IDs, students can enrol in no more than 3 courses per term)

> 📖 Try it yourself: tutorial Q4

Entity-Relationship (ER) Data Modelling

ER has three major modelling constructs:

▼ **Attribute**

Data item describing a property of interest

▼ **Entity**

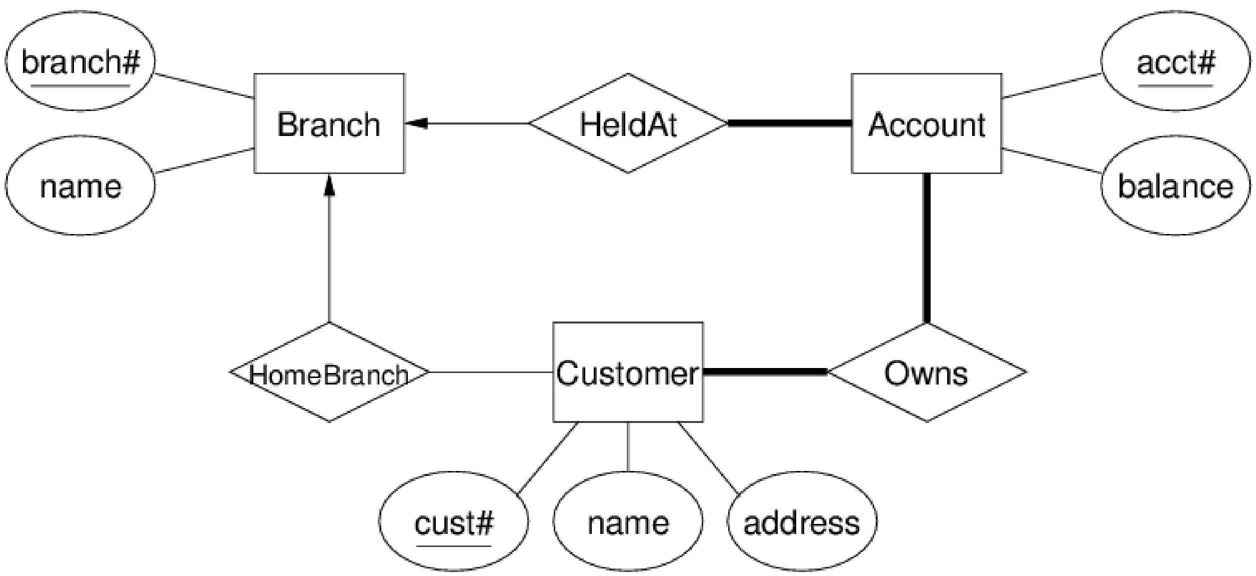
Collection of attributes describing object of interest

▼ **Relationship**

Association between entities (objects)

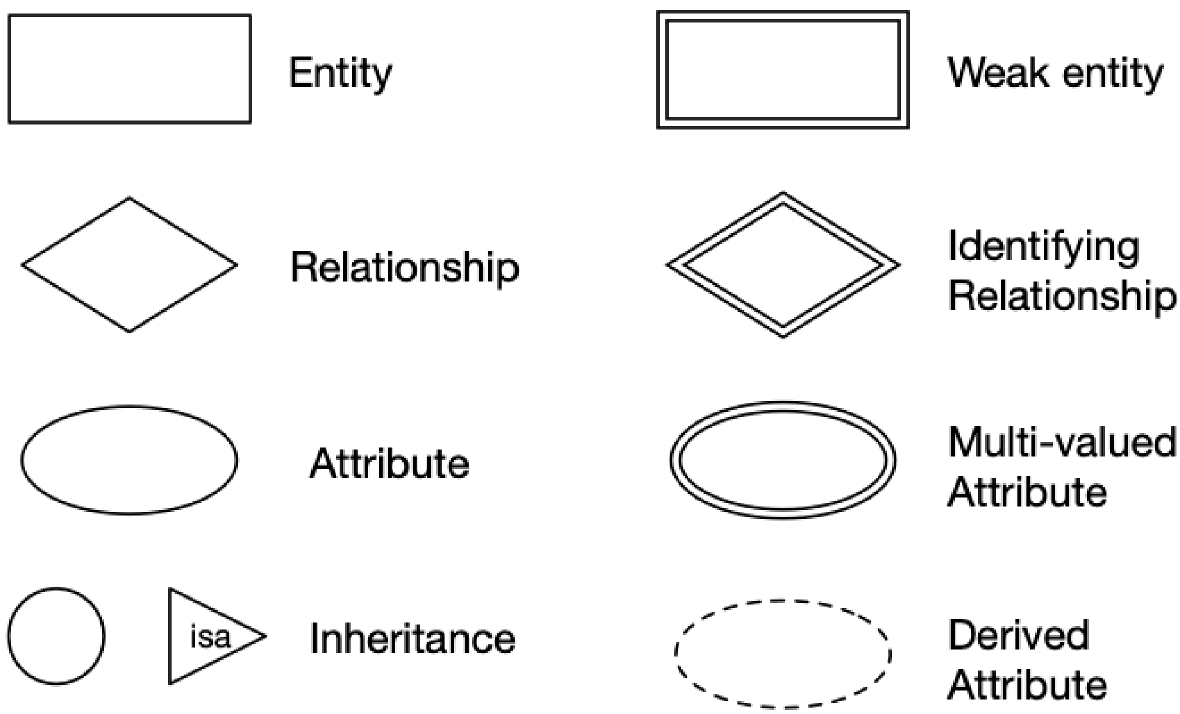
> 👆 Attributes are atomic values. Entities are collections of attributes.

Example: Banking system



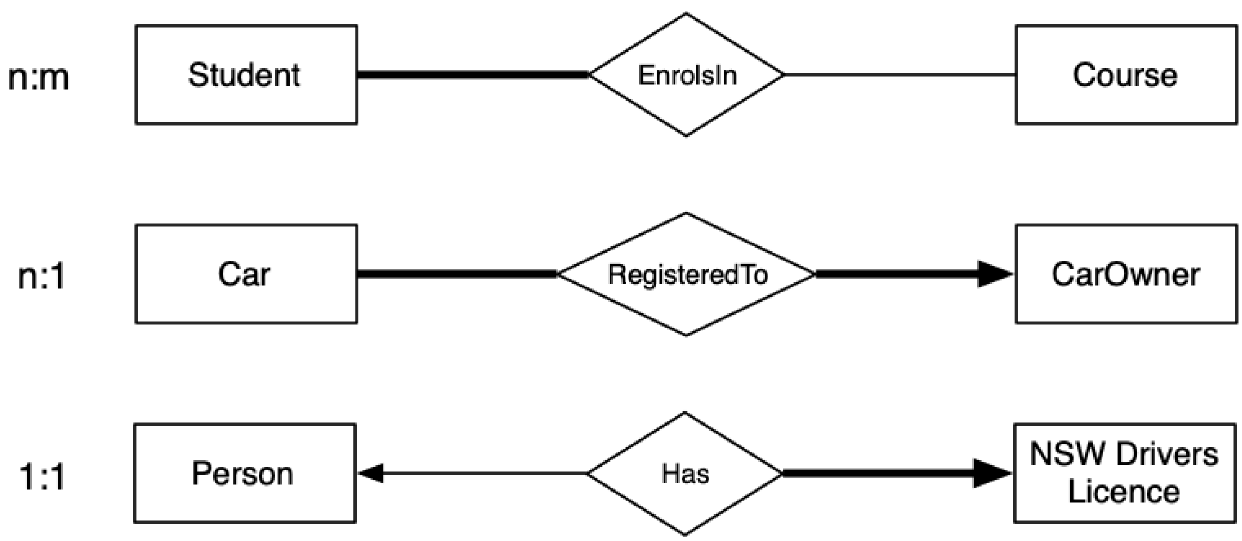
Underline: Primary key

▼ ER design elements



Niche things to know: partial keys (discriminators) in weak entities, difference between multi-valued attribute and making it an entity itself

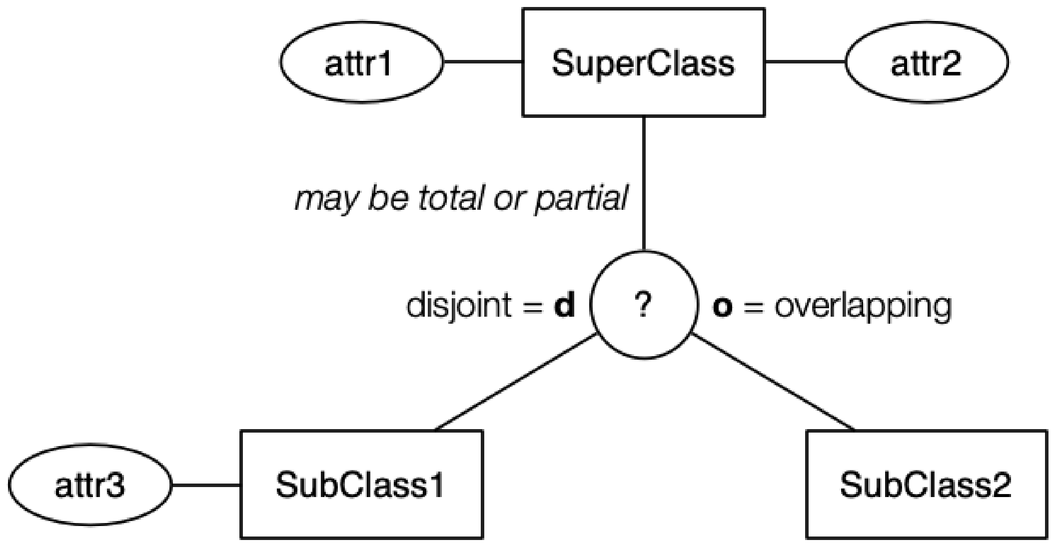
ER Relationships (totality and cardinality)



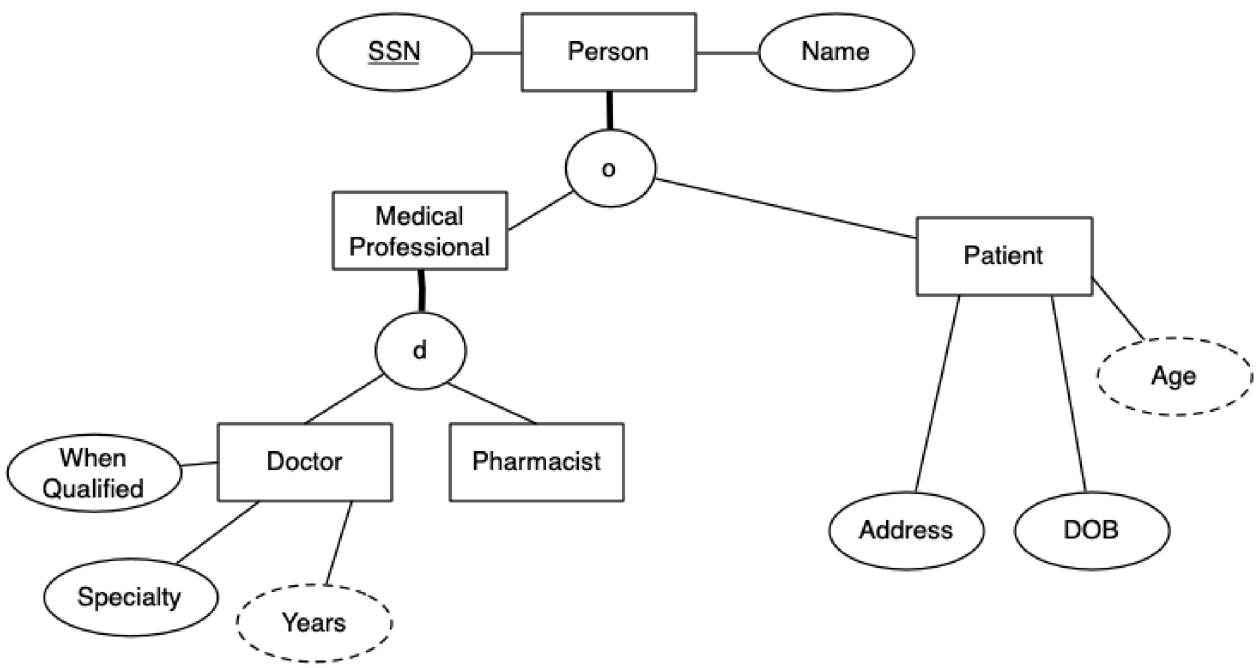
Thick line = total participation (at least 1); thin line = partial participation; arrow = at most 1

📖 Try: tutorial Q7, Q8, Q9

ER Class Hierarchies



Example: Medical Information



Relational Data Model

A collection of inter-connected **relations** (that look awfully close to tables)

Each **relation** (denoted R,S,T,...) has:

- a <u>name</u> (unique within a given database)
- a set of <u>attributes</u> (which can be viewed as column headings)

Each **attribute** (denoted A,B,... or a_1, a_2,...) has:

- a <u>name</u> (unique within a given relation)
- an associated <u>domain</u> (set of allowed values)

Relation: R($a_1 : D_1, a_2 : D_2, \ldots, a_n : D_n$)

Relational schema: a collection of relation schemas

Tuple of R: an element of $D_1 \times D_2 \times \ldots \times D_n$ (i.e. list of values)

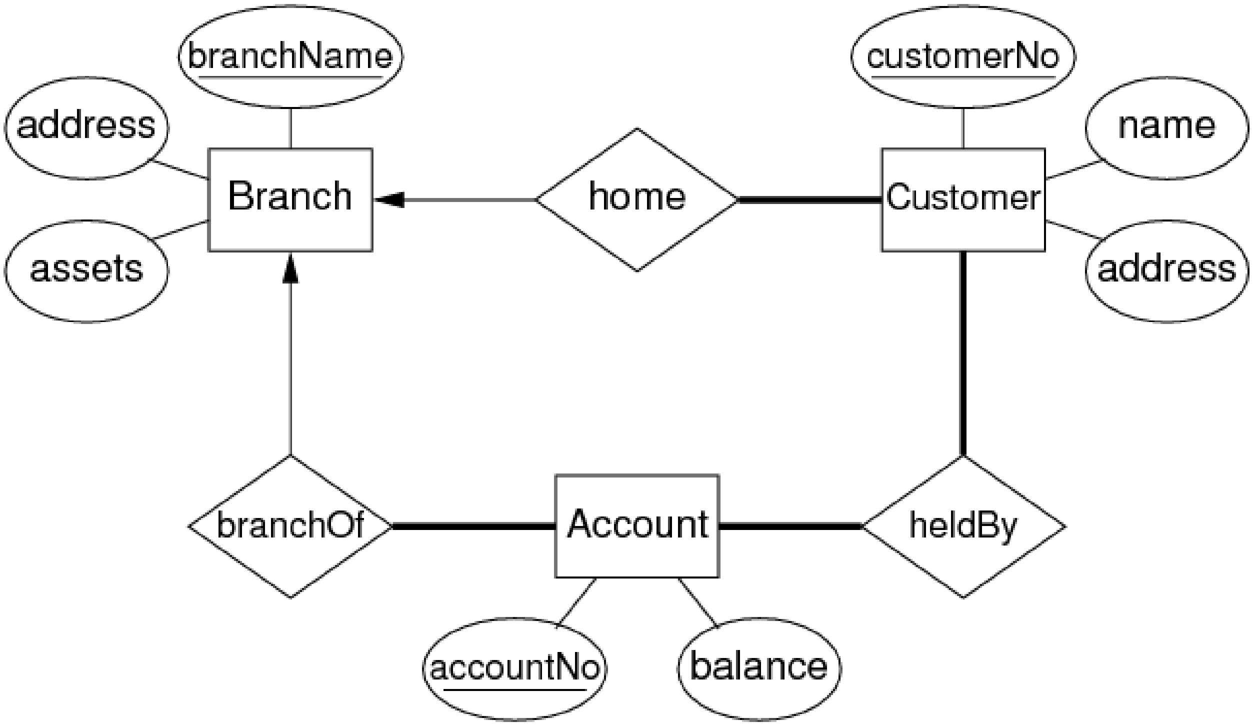
Instance of R: subset of $D_1 \times D_2 \times \ldots \times D_n$ (i.e. set of tuples)

<u>Example</u>: Bank Account

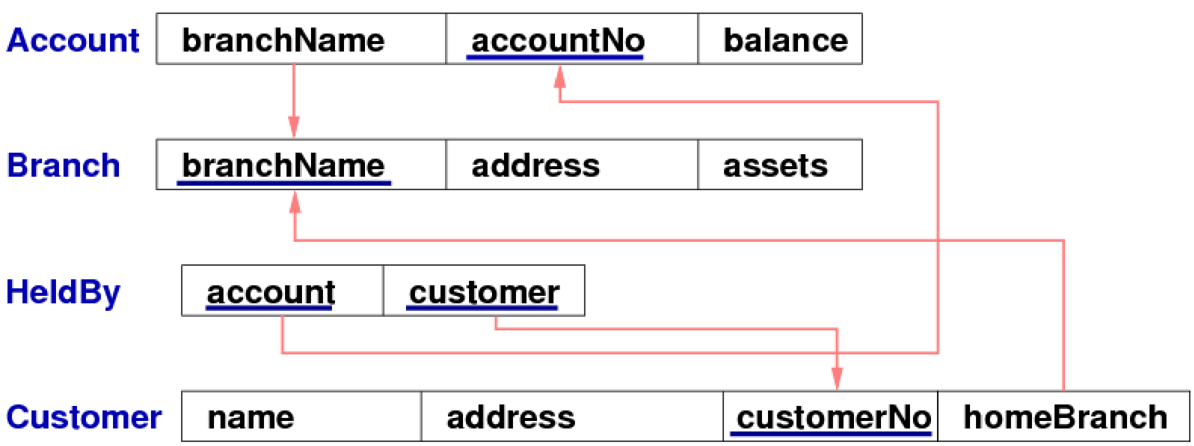
A relation: **Account(branchName, <u>accountNo</u>, balance)**. An *instance* of this relation:

```
{
  (Sydney, A-101, 500),
  (Coogee, A-215, 700),
  (Parramatta, A-102, 400),
  (Rouse Hill, A-305, 350),
  (Brighton, A-201, 900),
  (Kingsford, A-222, 700)
  (Brighton, A-217, 750)
}
```

ER → Relational



▼ Relational. **Identify the Primary and Foreign Keys**



Main takeaway this week: Relations, Relational schema, primary and foreign keys

SQL DDL

SQL data definition language (DDL) is the formal way of describing the above relational schemas. The primary SQL DDL construct is table creation

```sql
create table TableName (
    attr1Name  type [constraints],
    attr2Name  type [constraints],
    attr3Name  type [constraints],
    ...
    primary key (attrxName ),
    foreign key (attryName)
                references OtherTable (attrzName )
);
```

Let’s apply what we’ve learnt!

📖 11, 14, 15